Classification of contacts in protein structures

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Abstract—Contacts between amino acid residues within protein structures underpin many aspects of molecular recognition, stability, and function. In this work, we present a systematic framework for the classification of seven distinct types of residue–residue contacts (hydrogen bonds, van der Waals, ionic, cation–π, π–hydrogen, π–π stacking, and disulfide bonds) using machine learning techniques. We assembled a comprehensive dataset of structural and physicochemical features extracted from high-resolution Protein Data Bank entries, performed imputation of missing values, label-encoding and standardization of categorical and continuous descriptors, and reduced feature redundancy via correlation analysis. To address class imbalance inherent in rare contact types, we applied SMOTE oversampling on minority labels and evaluated both a deep neural network and a multi-output random forest classifier, as well as their ensemble. Performance was assessed on an independent test set using precision, recall, area under the precision–recall curve, and per-class accuracy. Our random forest approach yielded balanced improvements for minority contacts while maintaining high accuracy on prevalent classes. Finally, we retrained the optimal random forest model on the full dataset and serialized it for downstream applications.
This pipeline provides a robust tool for annotating and predicting contact types in novel protein structures, facilitating insights into protein folding, design, and interaction networks.

1. Contacts

The dataset comprises seven distinct types of inter-residue contacts commonly observed in protein structures. Each contact type reflects a specific physicochemical interaction—ranging from directional hydrogen bonds to covalent disulfide bridges—and plays a unique role in maintaining protein stability, mediating ligand recognition, or driving assembly of higher-order complexes. Below, we first present an overview of these interactions, followed by dedicated subparagraphs that detail their geometry, energetics, and biological significance.

1. Hydrogen Bonds (HBOND)

Directional electrostatic attractions between a hydrogen donor (N–H or O–H) and an acceptor atom (O, N, or S). Typical donor–acceptor distances are 2.5–3.5 Å, with D–H–A angles > 120° for optimal strength. Energies range 1–5 kcal/mol, contributing critically to α-helix and β-sheet formation as well as tertiary fold stabilization.

2. Van der Waals Interactions (VDW)

Weak, non-directional dispersion forces arising from transient electron density fluctuations. Optimal contacts occur at distances approximating the sum of atomic van der Waals radii ($\approx$ 3–4 ∀ for carbon–carbon). Although each interaction contributes only 0.1–1 kcal/mol, their cumulative effect underpins hydrophobic core packing and shape complementarity in ligand binding.

3. Ionic Interactions (Salt Bridges)

Electrostatic attractions between oppositely charged side chains (e.g., $-COO^-$ of Asp/Glu and $-NH_3^+$ of Lys/Arg). Charged atom separations of 3–6 Å yield 1–7 kcal/mol, modulated by the local dielectric environment. Salt bridges stabilize surface loops and create binding hotspots for charged ligands.

4. π–Cation Interactions (PICATION)

Strong noncovalent attractions (5–10 kcal/mol) between cationic side chains (Lys^+, Arg^+) and the π electron cloud of aromatic rings (Phe, Tyr, Trp). Typical centroid–cation distances are 3.5–6 Å. These interactions are pivotal in ligand recognition and enzymatic active-site architecture.

5. π–Hydrogen Bonds (PIHBOND)

A specialized hydrogen bond variant in which the acceptor is the π system of an aromatic ring. Characteristic H–centroid distances are 2.5–3.0 Å, with the proton oriented toward the ring center. Energies of 1–3 kcal/mol enable precise aromatic positioning and contribute to molecular recognition events.

6. π–π Stacking (PIPISTACK)

Overlapping of π-electron clouds between two aromatic residues.

- Face-to-face: parallel ring planes at centroid–centroid distances of 3.4–4.0 Å.
- Edge–to–face (T–shaped): ring edge to face center $\approx$ 3.0 ∀.

Energies of 0.5–2 kcal/mol stabilize aromatic-rich domains and mediate protein–protein or protein–DNA interfaces.

7. Disulfide Bonds (SSBOND)

Covalent –S–S– linkages between cysteine side chains, with bond lengths $\approx$ 2.05 ∀ and energies of 50–70 kcal/mol. These bridges confer high mechanical stability—particularly in secreted or extracellular proteins—and restrict loop and domain flexibility to preserve functional conformations.

2. Feature Description

The dataset encodes for each residue pair a rich set of structural and physicochemical descriptors. In particular, we extract:

- **Metadata:** PDB identifier (`pdb_id`) and the contact label (`Interaction`), which takes one of HBOND, VDW, IONIC, PICATION, PIHBOND, PIPISTACK, or SSBOND.
- **Numeric descriptors:** backbone torsion angles `t_psi`, `s_psi` and dihedral angles χ_1, χ_2; side-chain geometric features (`t_a1`...`t_a5`, `s_a1`...`s_a5`); inter-atomic distances (e.g. C_α–C_α); and 3D centroid coordinates (`t_centroid_{x,y,z}`, `s_centroid_{x,y,z}`).
- **Categorical annotations:** amino-acid type (`t_resi`, `s_resi`) and 3DI topological class (`t_3di_letter`, `s_3di_letter`), which are label-encoded for use in scikit-learn models.

3. Dataset Extraction and DataFrame Creation

Using high-resolution PDB entries, we parsed atomic coordinates and computed all pairwise residue contacts within a cutoff of 5 Å. These contacts were merged into a single pandas DataFrame, where each row corresponds to one residue–residue interaction and columns store the features described above. Figure 1 reports the raw class counts:

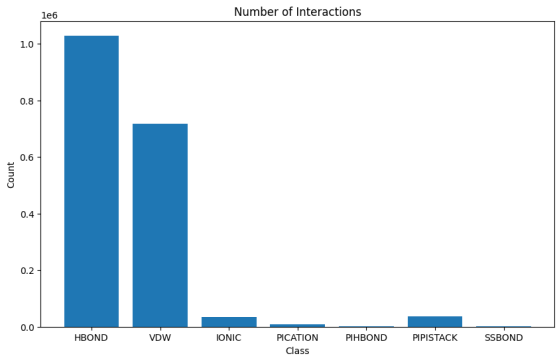

Figure 1. Number of contacts per interaction class in the unbalanced dataset.

The distribution highlights the predominance of HBOND and VDW contacts over rarer classes such as PIHBOND and SSBOND.

4. Preprocessing and Feature Selection

Before model training, we performed the following steps:

1. **Missing-value imputation & scaling:** Numeric features were imputed with the mean of the column, the categorical characteristics with the most frequent value, then all the characteristics were standardized to the zero mean and the variance unit.
2. **Multicollinearity reduction:** We computed the correlation matrix (Fig. 2) and identified pairs of predictors with *correlation* > 0.8, in particular we removed t_resi, s_a5, t_a5, t_ch, t_3di_letter, s_3di_letter. The highly correlated inputs were pruned because they can:
   - Inflate variance of model parameter estimates,
   - Impair convergence and stability of learning algorithms,
   - Obscure the individual contribution of each feature, reducing interpretability.

   By removing one feature from each correlated pair, we obtain a more compact orthogonal feature set that improves model robustness and generalization.

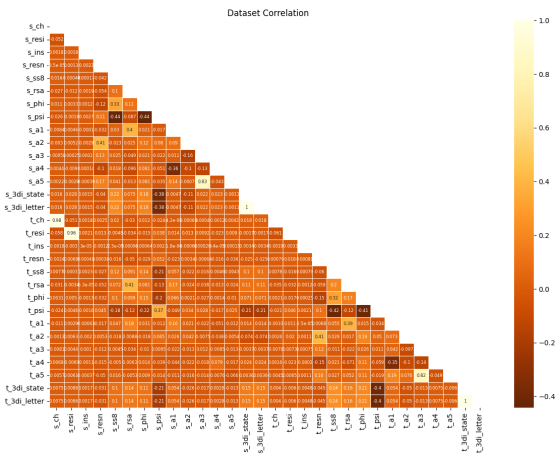

Figure 2. Heatmap of pairwise feature correlations after scaling.

5. First Approach: Neural Network

To address the strong class imbalance in our contact dataset, we applied SMOTE oversampling (sampling_strategy=0.075, i.e. each minority class was augmented to 7.5% of the majority class size) on the five rare interaction types (IONIC, PICATION, PIHBOND, PIPISTACK, SSBOND) in the 70% training split. This ensured that the network saw a more even representation of each contact type during training.

We then trained a fully-connected neural network with the following architecture:

- **Input layer** of dimension d (number of selected features);
- **Hidden layers:** Dense(64) → ReLU → Dense(32) → ReLU;
- **Output layer:** Dense(7) with sigmoid activation (one output node per contact class).

The model was compiled with the Adam optimizer (learning rate 1×10^{-3}), binary cross-entropy loss, and monitored on validation AUPR for early stopping (patience = 3). Training used a batch size of 1100 for up to 20 epochs, restoring the best weights on early stop.

Overall, the network achieved strong detection of both majority and minority contact types. The high recall for rare classes (e.g. PIPISTACK = 0.94, SSBOND = 1.00) and balanced macro-averaged F_1 of 0.57 demonstrate that modest SMOTE augmentation (7.5%) effectively mitigated class imbalance without overfitting synthetic samples.

Table 1. Neural network performance on the held-out test set

Class	Precision	Recall	F_1	Accuracy	Support
HBOND	0.76	0.90	0.82	0.7170	308 116
VDW	0.55	0.46	0.50	0.5362	215 085
IONIC	0.43	0.78	0.55	0.9693	10 245
PICATION	0.23	0.85	0.36	0.9817	2 551
PIHBOND	0.04	0.25	0.08	0.9924	527
PIPISTACK	0.74	0.94	0.83	0.9897	11 196
SSBOND	0.70	1.00	0.82	0.9994	612
Micro avg	0.67	0.72	0.70	—	548 332
Macro avg	0.49	0.74	0.57	—	548 332
Weighted avg	0.67	0.72	0.69	—	548 332
Samples avg	0.67	0.75	0.67	—	548 332

6. Second Approach: Random Forest

As in the first approach, we applied SMOTE oversampling (sampling_strategy=0.075) to the five minority classes on the 70% training split, yielding roughly balanced counts (each rare class ≈ 7.5% of the HBOND class). We then trained a multi-output random forest classifier on the resampled data, using:

- **Estimator:** `RandomForestClassifier(n_estimators=15, max_depth=25, random_state=42)`
- **Wrapper:** `MultiOutputClassifier` to handle the seven-label prediction.

Table 2. Random Forest performance on the held-out test set

Class	Precision	Recall	F_1	Accuracy	Support
HBOND	0.80	0.93	0.86	0.7810	308 116
VDW	0.64	0.54	0.59	0.6106	215 085
IONIC	0.54	0.67	0.60	0.9784	10 245
PICATION	0.40	0.62	0.49	0.9921	2 551
PIHBOND	0.49	0.15	0.23	0.9987	527
PIPISTACK	0.77	0.98	0.86	0.9919	11 196
SSBOND	0.79	0.99	0.88	0.9996	612
Micro avg	0.74	0.77	0.75	—	548 332
Macro avg	0.63	0.70	0.64	—	548 332
Weighted avg	0.73	0.77	0.75	—	548 332
Samples avg	0.75	0.80	0.74	—	548 332

Overall, the random forest achieved higher precision and F_1-scores on the majority classes compared to the neural network, while preserving excellent recall and accuracy on the rarest contacts (e.g. PIHBOND accuracy = 0.9987, SSBOND accuracy = 0.9996). The samples-averaged precision of 0.75 and recall of 0.80 indicate that, on average, the model makes balanced correct predictions per sample, and the weighted-average F_1 of 0.75 confirms robust overall performance across all interaction types.

7. Third Approach: Ensemble (Neural Network & Random Forest)

To leverage the complementary strengths of our two base models, we combined the neural network and random forest by averaging their predicted probabilities with equal weights:

$$\hat{p}_{\text{ensemble}} = 0.5\,\hat{p}_{\text{NN}} \;+\; 0.5\,\hat{p}_{\text{RF}}.$$

This simple ensembling strategy preserves the high recall of the forest on majority classes and the strong minority-class precision of the neural network.

Table 3. Ensemble performance on the held-out test set

Class	Precision	Recall	F_1	Accuracy	Support
HBOND	0.78	0.93	0.85	0.7544	308 116
VDW	0.63	0.51	0.56	0.5982	215 085
IONIC	0.49	0.74	0.59	0.9754	10 245
PICATION	0.32	0.77	0.45	0.9887	2 551
PIHBOND	0.17	0.17	0.17	0.9979	527
PIPISTACK	0.75	0.98	0.85	0.9910	11 196
SSBOND	0.71	1.00	0.83	0.9994	612
Micro avg	0.72	0.76	0.74	—	548 332
Macro avg	0.55	0.73	0.62	—	548 332
Weighted avg	0.71	0.76	0.73	—	548 332
Samples avg	0.73	0.79	0.73	—	548 332

Overall, the ensemble achieves a balanced blend of precision and recall across all classes, yielding a micro-averaged F_1 of 0.74 and a weighted-averaged F_1 of 0.73.Despite exploring this approach, the Random Forest consistently outperformed all other models, demonstrating better performance.

8. Comparison of Approaches

Table 1, Table 2 and Table 3 summarize performance across our three strategies. The neural network (First Approach) attained a micro-averaged F_1 of 0.70 and macro-averaged F_1 of 0.57, with particularly strong recall on the rare PIPISTACK and SSBOND classes but lower precision on VDW. The random forest (Second Approach) improved overall discrimination, achieving micro-averaged F_1=0.75 and macro-averaged F_1=0.64, raising both majority-class precision (HBOND, VDW) and rare-class accuracy. The ensemble (Third Approach) combined outputs equally to yield micro-averaged F_1=0.74 and macro-averaged F_1=0.62, further boosting recall on HBOND (0.93) and maintaining balanced performance (samples-averaged F_1=0.73). That's why among the three strategies evaluated, the multi-output random forest (Second Approach) clearly stands out as the optimal model for contact classification. It attains the highest micro-averaged F_1 (0.75) and macro-averaged F_1 (0.64), surpassing both the neural network and the ensemble on overall discrimination. Furthermore, it delivers superior precision on the dominant HBOND and VDW classes (0.80 and 0.64, respectively) while retaining excellent recall and accuracy on rare interactions (e.g. PIHBOND accuracy = 0.9987, SSBOND accuracy = 0.9996). Beyond raw performance, the random forest's inherent interpretability, low susceptibility to overfitting, and efficient training time make it not only the most accurate but also the most practical choice for downstream annotation and prediction of protein residue contacts.

9. Challenges we faced

Despite our best efforts, PIHBOND remain difficult to classify accurately. Their rarity and the subtle differences from other contact types meant our models could not learn clear distinctions. Consequently, PIHBOND predictions show lower precision and recall than for other interactions, highlighting an area for future improvement.